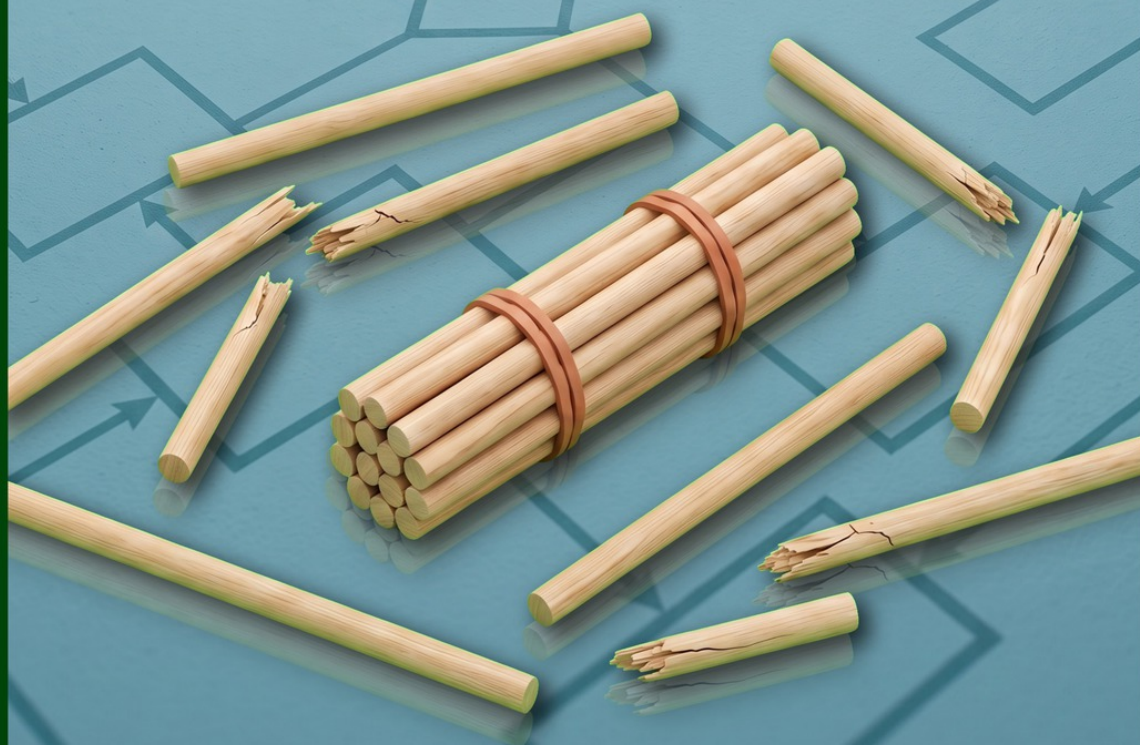


Antonio Ferrão Neto

THE METAPHOR OF THE BUBBLELAND

A PROTOSCIENTIFIC APPROACH

Appendices: Foundations and Hypotheses



Volume II

 **ImpNet**

THE METAPHOR OF THE BUBBLELAND

A Protoscientific Approach

Antonio Ferrão Neto

VOLUME II

Appendices: Foundations and Hypotheses



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About the cover

In unity there is strength. The saying describes, quite literally, the image on this cover: a firm **bound bundle of stalks**, surrounded by **broken twigs** scattered on the ground. The choice is deliberate, and it serves as a summary of what this volume — the one of **foundations and hypotheses** — is about.

The bundle is the central metaphor. Many of the conjectures in this book are, taken singly, fragile: none would stand on its own. But weak hypotheses, when *articulated* with one another, reinforce each other and come to sustain the whole — exactly like the stalks that, tied together, resist what would break each of them apart. This is the argument of the strengthening criteria **Cf1–Cf8**, developed in Appendix B (Types of hypotheses).

The broken twigs around it are the counterpoint: the hypotheses that do *not* stand, the ones that snap the moment they rest on themselves alone. The cover holds both fates side by side.

Every bundle, however, has a limit. Box 8, which marks where the bundle ceases to hold, points to that boundary — and it is there that we also bring in the etymology of the Latin *fascis*: the bundle of rods that, in ancient Rome, symbolized the authority of the magistrates. The word is raised deliberately. The correct reading of this cover is that of Aesop's fable — the bundle that does not break, though each rod breaks — and that of scientific *consilience*: the bundle is a figure of the strength that arises

from the union of weak parts, always presented together with its explicit limit, and not as an emblem.

In the background, a flowchart. It recalls both the systematics of the **scientific method**, treated in the first two appendices of this volume (Appendices A and B), and the many attempts to build efficient numerical methods for the “*stubborn*” problems discussed in Section V.1.

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Note to the reader

This is the second of the three volumes that make up *The Metaphor of the Bubbleland*. Whereas Volume I gathers the chapters of the work, this volume II, like volume III, is devoted exclusively to the appendices, bringing together the complementary material that supports the argument developed in the chapters.

Here the reader will find, in the first two appendices, a discussion of the scientific method and of how protoscience can be presented and discussed before becoming established science. Some appendices are merely documentary and present facts of established science; appendices P — The delayed-choice experiment, Q — The ultraviolet catastrophe and R — The photoelectric effect, for example, simply record well-known facts of Physics and serve to support some of the discussions developed in the text of the book. Other appendices, by contrast, are original and speculative, putting forward proposals for the development of new fields of science. Such is the case, for example, of appendices H — Topology and the fourth dimension in DNA, J — The external mind: points and counterpoints and K — The theory of biological evolution (this last one starts from the Theory of Evolution, a well-established fact in science, but proposes a new view: “Hyperevolution”). And there are also mathematical appendices, such as appendices E — A mathematical structure for Bubbleland, V — Types of “stubborn problems” in numerical computation and W — Numerical methods for ODEs/PDEs: a practical guide, which demonstrate that there are enor-

mous difficulties with the numerical methods currently available, and this justifies the use of the Successive Zooms Method, addressed in Volume III.

Throughout Volume I and Volume III, indications of the type “see Appendix . . .” refer directly to the sections gathered here. Such indications may also occur among the appendices themselves. Each appendix is identified by a letter and a title of its own, and can be located through the table of contents in the opening pages of the respective volumes.

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